

EXP-6

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Subject Name : Advanced Database Management System

A)

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

CODE:

```
CREATE TABLE Sales
(
    sale_date DATE,
    fruit VARCHAR(20),
    sold_num INT,

    PRIMARY KEY (sale_date, fruit)
);

INSERT INTO Sales
VALUES
('2020-05-01','apples',10),
('2020-05-01','oranges',8),

('2020-05-02','apples',15),
('2020-05-02','oranges',15),

('2020-05-03','apples',20),
('2020-05-03','oranges',0),

('2020-05-04','apples',15),
('2020-05-04','oranges',16);
```

```
select sale_date,  
sold_num - (select sold_num from sales  
            where fruit = 'oranges' and sale_date=S.sale_date)  
from Sales S  
where fruit='apples';
```

OUTPUT SCREENSHOTS:

	sale_date date	?column? integer
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

B)

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- **id** represents the current node.
- **p_id** represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

CODE:

```
CREATE TABLE Tree
(
    id INT PRIMARY KEY,
    p_id INT
);
```

```
INSERT INTO Tree
VALUES
(1,NULL),
(2,1),
(3,1),
(4,2),
(5,2);
```

```
select id ,case
when p_id is NULL then 'root'
when id in (select p_id from tree) then 'Inner'
else 'leave'
end as Type from Tree;
```

OUTPUT SCREENSHOTS:

	id [PK] integer 	type text 
1	1	root
2	2	Inner
3	3	leave
4	4	leave
5	5	leave

LEARNING OUTCOMES:

- Understand how hierarchical data is represented using parent-child relationships.
- Learn to classify nodes as Root, Inner, or Leaf using SQL conditions.
- Use CASE statements for conditional classification.
- Use subqueries to check whether a node has child nodes.
- Apply ORDER BY to display hierarchical results in sorted order.

CONCLUSION:

The SQL query successfully classified each node in the Tree table as Root, Inner, or Leaf based on its parent-child relationship and displayed the results in ascending order of node ID.